[INCH-POUND]

MIL-R-10509/8D
19 April 1989
SUPERSEDING
MIL-R-10509/8C
22 May 1984

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM (HIGH STABILITY),

STYLE RN50

INACTIVE FOR DESIGN AFTER 22 NOVEMBER 1968 (SEE 6.2). USE MIL-R-55182/7.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RN50, high-stability, film, fixed resistors. This style is available in characteristics C and E..

2. APPLICABLE DOCUMENT

* 2.1 Government document.

* 2.1.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-10509 - Resistors, Fixed, Film (High Stability), General Specification for.

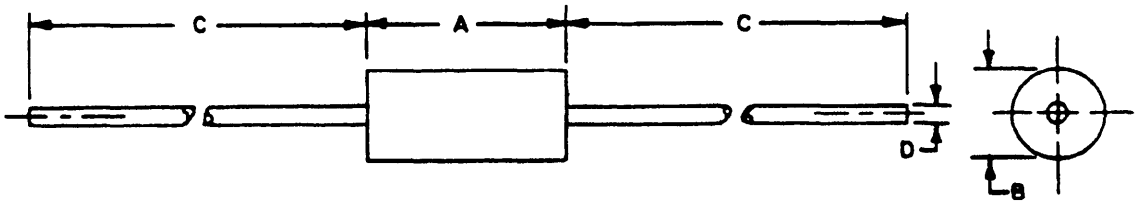
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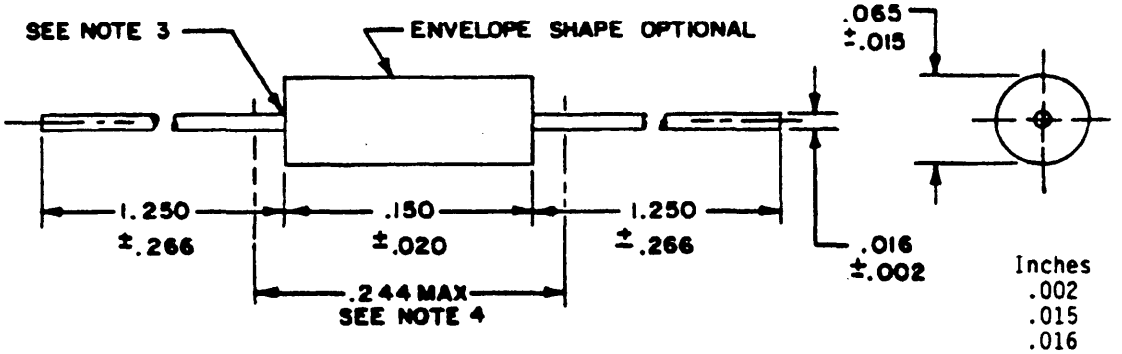
data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, Reliability, Logistics, and Standardization Division, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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Inches	mm
.002	0.05
.015	0.38
.016	0.41
.020	0.51
.062	1.65
.150	3.81
.244	6.20
.266	6.76
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
4. Maximum length is 'clean lead' to 'clean lead'..

* FIGURE 1. Style RN50 resistor.

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-10509, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/20 watt.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 200 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

a. Minimum resistance: Minimum resistance value for characteristics C and E shall be 10.0 ohms.

b. Maximum resistance: Maximum resistance value for characteristics C and E shall be 0.100 megohm.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-10509.

4.2 Terminal strength.

4.2.1 Direct load. Direct load shall be applied gradually until the load reaches 2 pounds.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements shall be in accordance with MIL-R-10509.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-10509 are applicable to this specification.

6.2 Substitution data. The resistors specified herein are not for use in design after 22 November 1968. They are authorized for use in design contracts effective prior to or on 22 November 1968. Existing equipment shall be supported by substitute resistors in MIL-R-55182/7 of the same resistance value, resistance tolerance, and characteristic performance of this specification regardless of failure rate designation. Resistors specified in MIL-R-55182/7 are preferred for design and regardless of failure rate designation can be used as substitutes for inactivated resistors of the same resistance value, tolerance, and characteristic performance of this specification. Characteristic substitution shall be made as indicated in table I.

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TABLE I. Characteristic substitution.

MIL-R-10509 characteristic	Substitute MIL-R-55182 characteristic
C	H
E	J

6.3 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1142)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-10509/8D	RESISTORS, FIXED, FILM (HIGH STABILITY), STYLE RN50

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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